

Large Language Models (LLMs): A Detailed Overview

Abstract

Large Language Models (LLMs) are advanced artificial intelligence systems trained on vast amounts of text data to understand and generate human-like language. They have become foundational to modern AI applications such as chatbots, translation systems, and content generation tools.

1. Introduction

LLMs are based on deep learning, particularly neural networks known as transformers. These models learn patterns, grammar, and context from massive datasets.

Examples include GPT models, BERT, and others. Their ability to generate coherent and context-aware responses makes them highly useful.

2. Architecture

Most LLMs use the Transformer architecture, which relies on self-attention mechanisms. This allows the model to focus on relevant parts of the input text.

Key components include: - Tokenization - Embeddings - Attention layers - Feed-forward networks

3. Training Process

LLMs are trained in two main phases:

Pretraining: Learning general language patterns from large datasets. Fine-tuning: Adjusting the model for specific tasks or domains.

Training requires significant computational resources and data.

4. Applications

LLMs are widely used in: - Chatbots and virtual assistants - Text summarization - Code generation - Language translation - Question answering systems

5. Limitations

Despite their capabilities, LLMs have limitations: - Hallucination (incorrect answers) - Bias in training data - High computational cost - Lack of real-time knowledge unless combined with systems like RAG

6. Conclusion

LLMs represent a major advancement in AI. With techniques like RAG and fine-tuning, their performance can be significantly enhanced for real-world applications.